

MILCAH MWANGI

Senior Software Engineer — Fintech Systems & AI Model Evaluation

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PROFESSIONAL SUMMARY

Software engineer with **20 years** building mission-critical backend systems for UK banks and fintechs (OakNorth, Close Brothers, Paragon Banking Group, Secure Trust Bank), progressing from IC to Principal Engineer leading teams of up to **12**, then moving into independent consulting and **4+ years** of contract AI model evaluation work. Delivered lending platforms processing **£150M+ in monthly commercial volume** at sub-120ms p99 latency, and scored **1,200+ model outputs** at a **95%+ QA acceptance rate** across Outlier, Telus, Alignerr, Scale AI, and Appen. Native-level English (C2). Fully remote, self-directed delivery across UK, US, and APAC time zones.

CORE COMPETENCIES

Languages & Frameworks: Python 3.x (expert), TypeScript, JavaScript, SQL, Java, FastAPI, React, Spring Boot

Financial Systems: SME credit-decisioning, loan-origination systems (LOS), risk engine integration, KYC/AML pipeline automation, FCA regulatory reporting, real-time settlement, Bacs/CHAPS integration

LLM Training & Evaluation: SFT prompt-response authoring, RLHF preference ranking, chain-of-thought reasoning review, rubric design, hallucination detection, multi-turn debugging scenario design, inter-annotator calibration

Testing & Code Review: pytest, TDD, PEP 8 enforcement, edge-case identification, pull-request governance, static analysis, API contract testing

Infrastructure & DevOps: Docker, AWS (ECS, Lambda, S3, RDS, CloudWatch), CI/CD (GitHub Actions, Jenkins), Kubernetes, Terraform, Linux

Data & ML: pandas, NumPy, SciPy, scikit-learn, TensorFlow, data pipeline orchestration, annotation QA (CVAT, Labelbox, Label Studio)

PROFESSIONAL EXPERIENCE

Senior Software Engineer — CodeSignal, Remote

Jan 2023 – Present

Python · FastAPI · PostgreSQL · AWS (ECS, Lambda, S3) · Docker · GitHub Actions · pytest

- Architected and shipped production FastAPI microservices powering NLP and AI evaluation workflows for enterprise clients, sustaining high uptime across PostgreSQL-on-AWS backends.
- Cut release cycle time from 9 days to 5 by building automated CI/CD pipelines in GitHub Actions, reducing manual deployment effort.
- Reduced production regressions by authoring and maintaining a comprehensive pytest suite covering unit, integration, and API contract tests, enforced as a merge gate.
- Reviewed pull requests across the engineering team for correctness, PEP 8 compliance, edge-case handling, and test fidelity against ticket specifications.
- Deployed containerized annotation-management and dataset-validation tools (Docker, AWS ECS, Lambda, S3) used daily across the organization.

Independent Consultant — Software Engineering & AI Evaluation, Remote

Aug 2018 – Dec 2022

Python · AI Model Training & Evaluation · Data Pipeline Engineering

- Delivered independent software engineering and data-pipeline consulting engagements for fintech and research clients following an 8-year tenure in full-time banking engineering roles.
- Built the foundation of a multi-year AI model training and evaluation practice during this period, contracting across Remotasks, DataAnnotation.tech, Handshake, Scale AI, and Appen — detailed in the AI Training & Evaluation section below.

Principal Software Engineer — OakNorth Bank, London, UK

Mar 2016 – Aug 2018

Python · Kafka · AWS Lambda · Open Banking APIs

- Led a team of engineers building automated SME credit-decisioning and loan-origination services processing £150M+ in monthly commercial lending.
- Reduced manual underwriting bottlenecks and cut time-to-decision from days to hours by designing event-driven microservices (Python, Kafka, AWS Lambda) for real-time credit-risk scoring.
- Led FCA compliance and Open Banking API integrations across core lending services, passing external regulatory audits with zero critical findings.
- Mentored mid-level engineers and established code-review standards that reduced post-release hotfixes.

Lead Software Engineer — Close Brothers Group, London, UK

Jan 2013 – Feb 2016

Python · Java · Oracle · JUnit · pytest

- Architected and delivered an invoice-finance risk-calculation engine processing 40,000+ SME ledger entries daily.
- Led a team of 12 developers and 3 QAs through a major core-banking migration with zero unplanned downtime.
- Introduced automated regression suites (JUnit, pytest) that reduced UAT defect leakage.
- Delivered bespoke regulatory reporting modules ahead of the FCA deadline, working directly with compliance and front-office teams.

Senior Software Engineer — Paragon Banking Group, London, UK

Jun 2009 – Dec 2012

Python · SQL · Tibco

- Built and maintained mortgage-underwriting and AML screening pipelines processing 12,000 buy-to-let applications/month, reducing false-positive alerts while preserving regulatory detection rates.
- Delivered a fraud-detection rule engine that flagged £4.1M in suspicious applications in its first quarter, earning the company's internal "Technology Excellence" award (2011).

Software Engineer — Secure Trust Bank, London, UK

Jul 2006 – May 2009

Java · Python · Oracle

- Developed core-lending and consumer-credit transaction services processing £400M in annual disbursements.
- Automated batch reconciliation workflows, cutting the overnight processing window from 5 hours to 1.8 hours.

AI TRAINING & EVALUATION EXPERIENCE

Contract work undertaken during the independent consulting period above (2018–2022) and, from 2023, alongside full-time employment at CodeSignal.

Code & Model Output Evaluator — Outlier, Telus Digital, Alignerr, Remote

2023 – 2025, part-time

Python · JavaScript/TypeScript · SFT · RLHF Preference Ranking

- Reviewed **300+ AI-generated code responses** (Python, JavaScript/TypeScript) for correctness, instruction adherence, security practices, and debugging quality, including chain-of-thought reasoning on algorithms and multi-step technical problems; identified **14 systematic failure patterns** adopted into model-retraining criteria.
- Designed 45 "broken-code" and multi-turn debugging scenarios testing model ability to identify, explain, and fix runtime errors, logic bugs, and performance bottlenecks.
- Scored **1,200+ technical responses** against structured rubrics, sustaining a **95%+ QA acceptance rate** with reproducible written feedback; flagged 60+ tests that passed against incorrect or incomplete solutions.

Annotation QA Lead & Guideline Author — Scale AI, Appen, Remote

2022 – 2024

Python · JSON · CSV · COCO · YOLO

- Audited **5,000+ annotation records** (JSON, CSV, COCO, YOLO) across large-scale text and image labeling projects, enforcing policy-style guidelines on ambiguous and edge-case content.
- **Improved inter-annotator agreement from 0.72 to 0.86 Cohen's κ** through structured weekly calibration sessions for 8–12 reviewers and iterative guideline revisions.
- **Reduced recurring schema and labeling errors 30%** by building Python validation scripts that detect inconsistencies and label drift pre-delivery; authored 10+ decision rules standardising gray-area judgments across the reviewer pool.

AI Trainer & Data Labeler — Remotasks, DataAnnotation.tech, Handshake, Remote

2021 – 2022

SFT Prompt Engineering · LLM Evaluation

- Created **800+ SFT prompt–response pairs** and evaluation instructions for LLM training; assessed outputs for accuracy, hallucination, and task alignment.
- Produced structured written rationales for **500+ evaluations** with error categorisation, completing high-volume batches within 24–48-hour SLAs.

PROJECTS

Multi-Lingual Sentiment Analysis Platform

Python · FastAPI · React · TensorFlow · Docker · AWS

- Built an end-to-end multilingual sentiment-analysis platform processing **50K+ documents/day** through scalable FastAPI REST APIs, backed by a comprehensive pytest suite covering preprocessing, inference, and API contracts.
- Containerized workloads with Docker and deployed on AWS with CloudWatch monitoring and auto-scaling; built a React dashboard visualising model outputs for **30+ internal analysts**.

EDUCATION

PhD, Computer Science (Part-Time) — Brunel University London

2001 – 2005

MSc, Computer Science — University of Westminster

1999 – 2000

BSc, Computer Science — University of Westminster

1995 – 1998